

Project Report

Interactive Quiz Web Application

Overview

This project is a simple and effective quiz web app designed to help students practice and learn better. The main idea is to give students a platform where they can attempt quizzes and also save the important questions they find while studying. Later, they can use these saved questions to create their own tests and revise them anytime.

Problem Statement

Most quiz websites only allow students to attempt ready-made quizzes. Students often come across important questions in books or classes, but they don't have one place to store and practice them in test format. Because of this, they forget many important topics.

Objective

To build a web app that allows students to practice quizzes and create their own question bank for future revision.

System Description

The app has two main parts:

1. **Practice Quiz** – Students attempt quizzes created from the existing question sets.
2. **Add Question** – Students can add their own important questions with answers and explanations. These questions are stored directly in the browser using local storage, and can be used later to create custom tests.

The app tracks performance, saves data in the user's device, and makes revision easier.

Unique Features

1. **Personal Question Bank**
Students can save important questions inside the app without any login. The data stays saved in their device.
2. **Create Custom Tests**
Students can create quizzes using only the questions they added, helping with targeted revision.

3. Adaptive Difficulty

The app checks how the student is performing. If they perform well, difficulty increases. If they struggle, easier questions appear.

4. Revision Reminders

The app recommends when to revise questions based on spaced repetition to improve memory.

5. Performance Analytics

Students can view accuracy, improvement, weak topics, and strong areas through a simple dashboard.

6. Question Sharing (Optional)

Students can export selected questions and share them with friends.

7. Explanation Support

Each question can include a detailed explanation for better understanding during revision.

8. Offline Support

The app works even without internet because all data is saved locally in the browser.

Technologies Used

Frontend: HTML, CSS, JavaScript

Storage: Browser LocalStorage

Backend: Not required for basic features

Workflow

1. User opens the web app.
2. Chooses between taking a quiz or adding new questions.
3. In practice mode, the quiz loads and answers are checked instantly.
4. In add-question mode, the student enters a question, options, correct answer, and explanation.
5. The app stores all data in the browser.
6. Students can generate custom quizzes any time using their saved questions.
7. Analytics update automatically after every quiz.

Conclusion

This project stands out because it is not just a quiz app. It allows students to build their own question bank, practice it anytime, and revise smartly. With no login required and

complete offline support, the app becomes a fast, accessible, and reliable study companion.